



EDA AI LAB 12 Tasks

TASK: EXPLORATORY DATA ANALYSIS AND PRICE PREDICTION OF HOUSES

- Perform EDA (Exploratory Data Analysis)
- Handle missing values
- Analyze feature relationships
- Build a basic regression model to predict house prices

1. Data Understanding

- a) Load the dataset using Pandas
- b) Display:
 - First 5 rows
 - Dataset shape
 - Column names
- c) Identify:
 - Numerical vs Categorical features

2. Data Cleaning

- a) Find missing values
- b) Handle missing data:
 - Numerical → fill with mean/median
 - Categorical → fill with mode
- c) Drop columns with too many missing values (e.g., >50%)

3. Univariate Analysis

- a) Plot distribution of:
 - SalePrice
 - GrLivArea
- b) Identify skewness and outliers

TOOLS:

- Histogram
- Boxplot

4. Bivariate Analysis

- a) Find correlation with SalePrice
- b) Plot:
 - Scatter plot: GrLivArea vs SalePrice
 - Bar plot: OverallQual vs SalePrice
- c) Identify top 5 features affecting price



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5. Multivariate Analysis

- a) Create a correlation heatmap
- b) Analyze strong positive/negative relationships

6. Feature Engineering

- a) Create new feature:
 - $\text{HouseAge} = \text{YrSold} - \text{YearBuilt}$
- b) Drop irrelevant features (like Id)

7. Model Building

- a) Split data into training/testing
- b) Apply:
 - Linear Regression
- c) Evaluate model using:
 - MAE
 - RMSE
 - R^2 Score